## QUICK REFERENCE

| KEY | IN THE TERMINAL |
|---|---|
| BREAK | Menu |
| CAPS + 1 | Esc |
| CAPS + 2 | Caps lock |
| CAPS + 3 | Tab |
| CAPS + 5–8 | Arrows |
| CAPS + 9 | 32 / 64 / 80 |
| CAPS + 0 | Backspace |
| CAPS + SYM, key | Ctrl-key |
| SYM + A S D | ~ \| \ |
| SYM + F G H | { } ^ |
| SYM + Y U X | [ ] ` |

| KEY | IN THE MENU |
|---|---|
| D | Phonebook |
| H | Hang up |
| W | WiFi |
| S | Settings (S saves) |
| X | XMODEM download |
| C | Clear the screen |
| Q | Quit to BASIC |
| BREAK | Back |

### DIALING BY HAND

`ATDTX"host:port"` telnet · `ATDX"host:port"` raw · `ATI` status · `ATW` scan WiFi

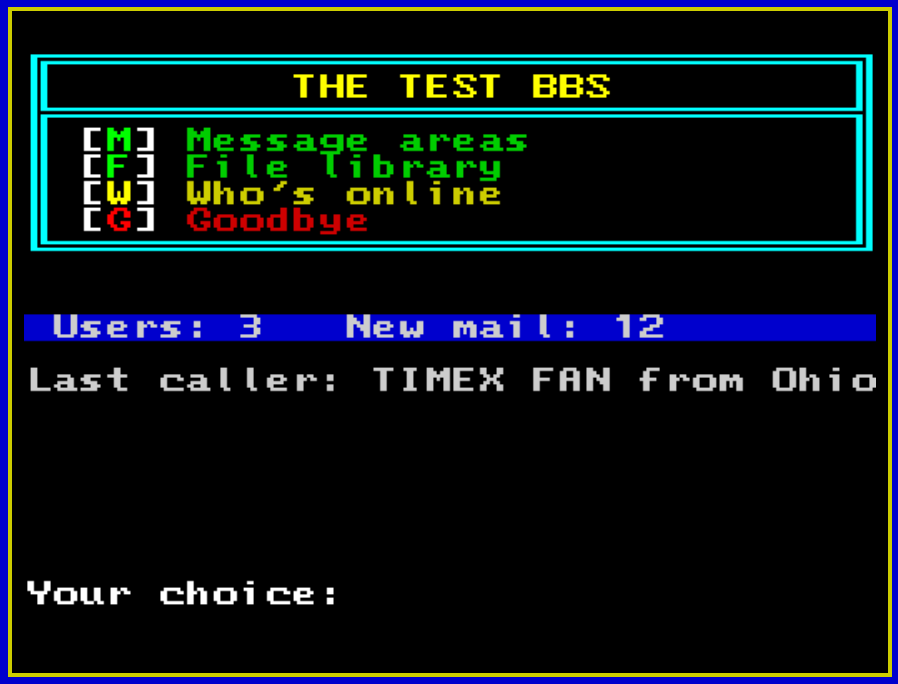
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TIMEX/SINCLAIR 2068 · TS-PICO V3 · 2026

# PICOTERM

## USER MANUAL

A terminal program for the TS-2068 and the TS-Pico v3 card. The card's WiFi modem runs Zimodem, and PicoTerm uses it to reach telnet BBSes: ANSI color and box drawing, 32 or 64 columns (80 with PicoVideo), a phonebook, and XMODEM downloads to the SD card.




TS-2068 + TS-Pico v3 · WiFi

Version 0.6, 2026

timexsinclair.com




Start-up: Zimodem answers each command with OK.

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**PicoTerm** 0.6 © 2026. Written in Z80 assembly for the TS-Pico v3 card.

Built on Bo Zimmerman's Zimodem, the TS-Pico firmware and ROM, and the TS 2068 Technical Manual's Video Mode Change Service (as Zebra's OS-64 uses it). The CP437 font is PicoVideo's: Bm437_ACM_VGA_8x8 from VileR's Ultimate Oldschool PC Font Pack (int10h.org), CC BY-SA 4.0.

This manual was written from the program itself. Its screenshots come from PicoTerm's own code run on a model of the 2068 with a simulated modem and BBS, so the BBS screens are examples.

`picoterm.tap: loader CLEAR 34815: LOAD "" CODE: RANDOMIZE USR 34816 · code at 8800h · 8251 $73/$77 · TS-Pico 14/15 · AY F5h/F6h`



# YOUR BBS LIST

| NAME | HOST:PORT | TELNET | COLS | NOTES |
|---|---|---|---|---|
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## 13 LIMITS

- **Downloads only, and at most 6,912 bytes**: no upload, YMODEM or ZMODEM, and no capture to a file.
- **No scrollback.** Text that scrolls off the top is gone.
- **24 rows.** BBSes that assume 25 lose the bottom line.
- **Color:** no bright background on its own; 64 columns on a stock 2068 has one color pair.
- **Keyboard:** the symbol keys follow the Spectrum's legends, which the 2068 mostly shares; the tables in section 4 are what PicoTerm does.
- **Screen width:** PicoTerm can't tell a telnet host your width itself. Most BBSes ask, or measure it with the cursor-position question, which PicoTerm answers.
- **The modem:** the TS-Pico's, in Hayes mode, with its 8251 emulation. Not a stock TS2050.
- **Ctrl-S and Ctrl-Q** don't reach the host on a call with flow control.

## 14 ABOUT THIS EDITION

PicoTerm 0.6, 2026, for the TS-Pico v3 card, in Z80 assembly (sjasmplus). This manual was written from the program itself; the BBS screens are examples from a simulated BBS, not real systems. Credits are inside the front cover.



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KEYS IN THIS MANUAL

`CAPS` is CAPS SHIFT and `SYM` is SYMBOL SHIFT. `BREAK` is `CAPS`+`SPACE`; it opens PicoTerm's menu, and in any PicoTerm screen it goes back one step.

# 01 WHAT YOU NEED

- **A TS-2068** with the TS-Pico ROM 2.3 or later.
- **The TS-Pico v3 card**, with firmware that has the TS2050 modem emulation (the 8251 at ports `$73` and `$77`), and Zimodem on the card's ESP32. The card's modem has to be in *Hayes mode*. PicoTerm switches it there itself where the firmware allows.
- **A WiFi network** the card can join. PicoTerm can scan for one and join it (section 9).
- **An SD card in the TS-Pico**, for `picoterm.cfg` (your settings and phonebook) and for downloads.
- *Optional:* **PicoVideo**, for 80 columns and for color in 64 columns, on its VGA output.

A REAL TS2050?

PicoTerm is written for the TS-Pico's modem. It doesn't pulse-dial, and it doesn't watch the TS2050's carrier line, so it won't drive an original 300-baud TS2050. Use MTERM for that.

# 02 LOADING PICOTERM

`picoterm.tap` holds a one-line BASIC loader and about 15 KB of machine code, which loads at 34816 (8800h).

1. Copy `picoterm.tap` to the SD card's `/TAP` folder.
2. Type `LOAD "tpi:picoterm.tap"` `ENTER` to mount it.
3. Type `LOAD ""` `ENTER`. The loader runs by itself.



# 12 MESSAGES AND PROBLEMS

| YOU SEE | IT MEANS, AND WHAT TO DO |
|---|---|
| `Modem ready.` | Zimodem answered. Type or dial. |
| `This TS-Pico has no tpi:modem.` | Older card firmware. Set the card's modem to Hayes mode yourself (see the TS-Pico manual), or update. |
| `No modem: the 8251 isn't ready.` | Nothing at `$73` / `$77` within 5 s: no v3 card, the modem off, or no modem firmware. PicoTerm runs; modem keys are dropped. |
| `No answer from the modem.` | No `OK` in about 12 s. Check the modem is in **Hayes** mode (TS2050 mode hides Zimodem's replies), then type `AT` `ENTER`. |
| `NO ANSWER`, `ERROR` | Zimodem couldn't connect, or didn't understand: check the host, port, quotes and WiFi (`ATI` shows Zimodem's state). |
| `NO CARRIER` | The host hung up. |
| `Hang up first.` | Dialing or WiFi while online: `BREAK`, `H`. |
| `Couldn't save the settings.` | No TS-Pico or SD card, or the card refused. |
| `Too big for memory (6912).` | The download was too long for this version. |
| `The sender didn't start.` | Nothing from the BBS for about 30 s. Start the download on the BBS first. |
| `Too many errors. Canceled.` | Ten bad blocks in a row. Try again. |

SLOW TEXT

The card passes text to the 2068 no faster than its `MODEM_BAUD` setting: 300 baud unless changed, 30 characters a second. Raise it in the TS-Pico's `config.ini`: `0` (no limit) is safe with flow control. Without it, keep to about 2400, or a fast host can overrun the card.

TEXT MISSING IN A FAST STREAM

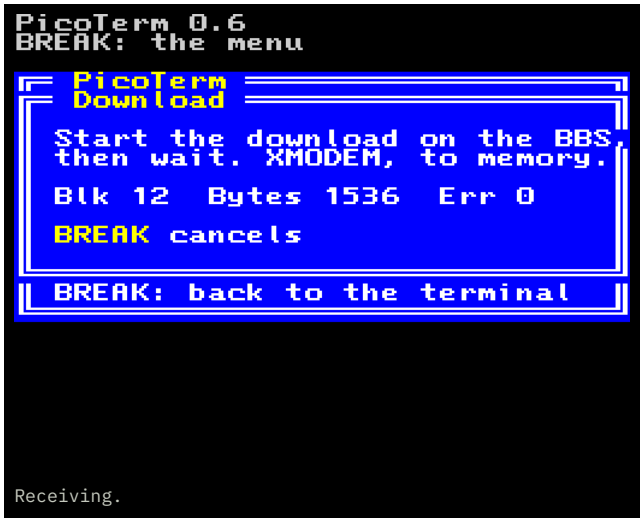
Dial with the `X` (`ATDTX`, or the phonebook), and check the card's firmware has `tpi:modem flow`.

# 11 DOWNLOADING FILES

PicoTerm receives with **XMODEM**, CRC or 1K, with checksums for older senders. The file goes to the TS-Pico's SD card.

1. On the BBS, choose the file and XMODEM.
2. `BREAK`, `X`, and a name for the file (`download.bin`). PicoTerm asks before replacing a file.
3. The window counts blocks, bytes and errors. Damaged blocks are asked for again.
4. **Strip ^Z padding?** XMODEM pads the last block with 1Ah. `Y` for text; `N` for programs and binaries.
5. PicoTerm saves the file and shows its size.

`BREAK` cancels and tells the BBS. In stock 64-column mode, PicoTerm uses 32 columns during the transfer: the file is held in the 64-column screen's memory.




IN THIS VERSION

Downloads are received into memory first: at most **6,912 bytes**. The file is saved as `SAVE "f:name" CODE` saves it, so it starts with a 128-byte +3DOS header. `LOAD "f:name" CODE` loads it back; on a PC, cut the first 128 bytes. A planned TS-Pico firmware addition should lift both limits.



## WHAT HAPPENS AT START-UP

1. PicoTerm reads `picoterm.cfg` from the card's current folder, if there is one, and sets up the screen in its column mode.
2. It puts the card's modem in Hayes mode (`tpi:modem hayes`).
3. It resets the 8251, which starts the card's ESP32.
4. It sends `AT` once a second until Zimodem answers `OK` (up to about 12 s; Zimodem's banner may show first), then your init string (`ATE1V1X1Q0`) and the telnet terminal type (`AT&S"41=ANSI"`).
5. It turns on flow control: `ATF1`, then `tpi:modem flow on`. Older firmware refuses, and PicoTerm carries on without.
6. It shows `Modem ready.` You see everything the modem says. If something is missing, a message says so and PicoTerm carries on (section 12).

HAYES MODE STAYS

The TS-Pico saves the modem's mode in `config.ini`, so the card stays in Hayes mode after you quit. Before loading MTERM or another TS2050 program, type `SAVE "tpi:modem 2050"`.

MEMORY

The loader does `CLEAR 34815`. PicoTerm also opens the second display file with the ROM's Video Mode Change Service, as OS-64 does, which moves BASIC up to 7B00h. It stays open after you quit, so BASIC has about 7 KB less room until you reset.

## 03 YOUR FIRST CALL

Once the modem is ready, everything you type goes to it. Zimodem echoes your typing and answers in words ( `OK` , `CONNECT` , `NO CARRIER` ), like a Hayes modem.

1. **Join your WiFi**, once: `BREAK`, then `W` (section 9).
2. **Dial a host** by typing its address in quotes: `ATDTX"bbs.example.com:23"` `ENTER`. The `T` asks for telnet, which most BBSes want, and the `X` turns on flow control. `ATDX"…"` is a raw TCP port.
3. Zimodem answers `CONNECT`. From now on everything you type goes to the BBS.
4. **To hang up:** `BREAK`, then `H`. If the BBS hangs up first, you see `NO CARRIER`.

Put the hosts you use in the phonebook (section 8) and dial them with one key; it adds the `X` itself.

WHY THE X

When the BBS sends faster than the 2068 draws, the card tells Zimodem to pause and go on (XOFF, XON). Zimodem obeys only on a call dialed with `X` ; without it a fast host can overrun the card. It needs firmware with `tpi:modem flow` .






## 10 SETTINGS

`BREAK`, then `S`. Each key changes one setting at once.

| KEY | SETTING | DEFAULT |
|---|---|---|
| `E` | Emulation | ANSI |
| `C` | Columns at start-up | 32 |
| `P` | PicoVideo fitted | No |
| `I` | 64-column ink | White |
| `B` | Backspace: BS or DEL | BS |
| `L` | Local echo | No |
| `N` | ENTER sends CR or CR LF | CR |
| `F` | Blink shows as FLASH | Yes |
| `G` | Bell | Yes |
| `M` | Modem init string | `ATE1V1X1Q0` |
| `S` | **Save** settings and phonebook | |

### PICOTERM.CFG

`S` writes `picoterm.cfg` in the SD card's current folder; PicoTerm reads it at start-up. It's plain text: `emul=2` (0 TTY, 1 VT52, 2 ANSI), `cols=1` (0 is 32, 1 is 64, 2 is 80), `picovideo` , `ink64` , `bs` (8 or 127), `echo` , `enter` , `blink` , `bell` , `init` , and a line per phonebook entry:

```
phone=Test BBS|bbs.example.com:23|1|2|0|8|0
```

That's the name, host, telnet, emulation, columns, backspace and echo. Lines PicoTerm doesn't understand are skipped.

## 09 WIFI

BREAK, then W, while offline. Zimodem scans ( ATW ) and PicoTerm lists up to 16 networks.

1. Choose one with the arrows and press ENTER, or M to type a name that isn't listed. R scans again.
2. Type the password (it shows as stars) and press ENTER. PicoTerm sends `ATW"name,password"` and waits up to 20 s.
3. If it worked, Y makes it the default ( AT&W ): Zimodem joins it by itself from then on.



The networks Zimodem can see.

COMMAS AND QUOTES

Zimodem's `ATW"name,password"` can't carry a comma or a `"` in either part.



## 04 THE KEYBOARD

Letters, digits and SPACE type as you'd expect; CAPS gives capitals. Every other ASCII character and control key is on a shifted key. A held key repeats after about half a second.

| KEYS | LEGEND | SENDS OR DOES |
|---|---|---|
| CAPS+1 | EDIT | **Esc** (1Bh) |
| CAPS+2 | CAPS LOCK | Caps lock, for letters |
| CAPS+3 | TRUE VID | **Tab** (09h) |
| CAPS+5–8 | ← ↓ ↑ → | **Arrows:** `ESC [ D B A C` (ANSI, TTY), `ESC D B A C` (VT52). In PicoTerm's screens they move the selection. |
| CAPS+9 | GRAPHICS | **Next column mode:** 32, 64, 80 |
| CAPS+0 | DELETE | **Backspace:** BS (08h) or DEL (7Fh), as set |
| BREAK | BREAK | **The PicoTerm menu**; in a menu, back |
| ENTER | | CR, or CR LF as set |

### CONTROL KEYS

Press CAPS and SYM together, let go, then press a letter: PicoTerm sends it as a control key. CAPS+SYM, C sends **Ctrl-C**. Pressing CAPS+SYM twice cancels it. It works for the letters and [ \ ] ^ _. With flow control, Zimodem keeps Ctrl-S and Ctrl-Q for itself.

## SYMBOL SHIFT

The red symbols work as in BASIC. The keys that type keywords give the ASCII characters the 2068 can't otherwise type:

| KEY | SENDS | KEY | SENDS | KEY | SENDS |
|---|---|---|---|---|---|
| SYM+A | ~ | SYM+S | \| | SYM+D | \ |
| SYM+F | { | SYM+G | } | SYM+H | ^ |
| SYM+Y | [ | SYM+U | ] | SYM+X | ` (not £) |

| KEY | SENDS | KEY | SENDS | KEY | SENDS |
|---|---|---|---|---|---|
| SYM+Z | : | SYM+C | ? | SYM+V | / |
| SYM+R | < | SYM+T | > | SYM+O | ; |
| SYM+P | " | SYM+J | - | SYM+K | + |
| SYM+L | = | SYM+B | * | SYM+N | , |
| SYM+M | . | SYM+1…0 | ! @ # $ % & ' ( ) _ | | |

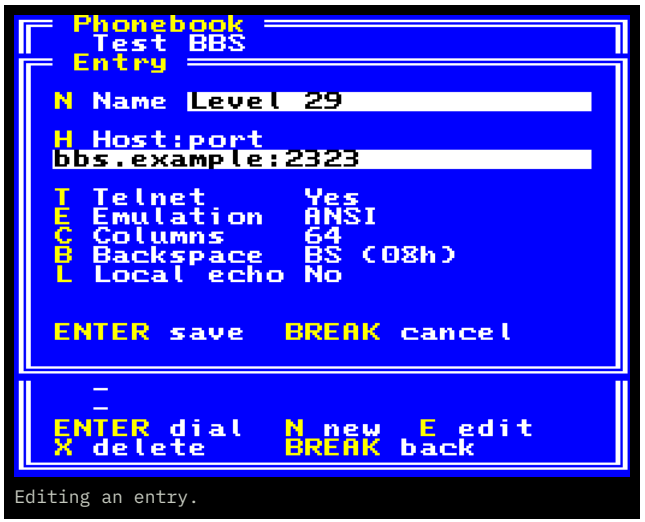
SYM with Q, W, E or I (≤ ≠ ≥ AT) sends nothing.



## AN ENTRY

| KEY | FIELD | NOTES |
|---|---|---|
| N | Name | Up to 16 characters |
| H | Host:port | Up to 40, e.g. `bbs.example.com:23` |
| T | Telnet | Yes: `ATDT`. No: `ATD`, raw TCP |
| E | Emulation | ANSI, VT52, TTY |
| C | Columns | 32, 64, 80 (64 without PicoVideo) |
| B | Backspace | BS or DEL, as the host wants |
| L | Local echo | For hosts that don't echo |

ENTER saves the entry; BREAK drops the changes. In a text field DELETE rubs out and ENTER accepts.




Editing an entry.

## 08 THE PHONEBOOK

20 hosts. Move with CAPS+6 and CAPS+7.

| KEY | DOES |
|---|---|
| ENTER | **Dial**: applies the entry's settings, goes back to the terminal, sends `ATDTX"host:port"` (`ATDX` with telnet off). |
| N | **New**: the name, ENTER, the host, ENTER, then ENTER to save. It starts with your current settings. |
| E / X | **Edit / delete** (Y confirms). |



The phonebook.

REMEMBER TO SAVE

Entries live in memory until you save your settings (Settings S). Dialing while online is refused: “Hang up first.”



## 05 THE TERMINAL

PicoTerm has three emulations. It uses ANSI unless you choose otherwise in Settings or in a phonebook entry, and it tells Zimodem which one to report to telnet hosts.

| EMULATION | WHAT IT UNDERSTANDS |
|---|---|
| **ANSI** | For BBSes: a VT100 subset with color, as BBS software expects of "ANSI-BBS".<br>• Cursor moves, positioning, save and restore.<br>• Erasing; inserting and deleting lines and characters.<br>• A scrolling region.<br>• Colors with bold, blink and reverse.<br>• It answers the cursor-position question, which BBSes use to find the screen size, and identifies itself as a VT100. |
| **VT52** | The standard set: cursor moves, home, `ESC Y` addressing, erasing, reverse line feed, identify. |
| **TTY** | Text and control codes only; escape sequences show as text. |

In every emulation PicoTerm handles **BEL**, a short tone on the AY; **BS**; **Tab**, every 8 columns; **LF**; **CR**; and **FF**, which clears the screen. Text wraps as on a VT100: a character in the last column leaves the cursor there, and the next starts a new line.

### COLOR AND CHARACTERS

ANSI's eight colors become the 2068's. **Bold** is BRIGHT, **blink** is FLASH (or not, in Settings), and **reverse** swaps ink and paper. BRIGHT lights the whole cell, so bold text on a colored background brightens the background too.

PicoTerm uses the PC's CP437 characters, as PicoVideo does for 80 columns, so box lines, shading and accented letters look as BBS authors meant.

## 06 32, 64 AND 80 COLUMNS

CAPS+9 changes mode, keeping what's on screen. Settings C picks the mode PicoTerm starts in.

| MODE | NEEDS | COLOR |
|---|---|---|
| **32** | any 2068 | Per character. Readable on any TV. |
| **64** | any 2068 | One ink and paper (Settings I); per character with PicoVideo. |
| **80** | PicoVideo | Per character, on VGA. The 2068's own video shows nothing useful. Turn on **PicoVideo** in Settings. |










The screen is 24 rows. A line scrolls in about 40 ms in 32 and 80 columns and 100 ms in 64: fast enough for 2400 baud.



## 07 THE MENU

Press BREAK at any time. The menu's top line says whether you're online, and the emulation and columns in use.

| KEY | DOES |
|---|---|
| D | The phonebook: dial, add, edit, delete |
| H | Hang up |
| W | WiFi: scan, join, save |
| S | Settings |
| X | Download a file (XMODEM) |
| C | Clear the screen |
| Q | Quit to BASIC; hangs up first if online |
| BREAK | Back to the terminal |

Text from the BBS waits while a menu is open, and appears when you return.

